

Importance-Weight Guidance in Block Diffusion — Rigorous Formulation

2026-07-17 · Companion to MASKED_GUIDANCE.pdf (absorbing-kernel instantiation, Eqs. 1–5 therein) · Implementation: models_seq/bd_models.py :: plan_guided / _denoise_block_mask_guided, models_seq/bd_disc.py, train_bd_disc.py, eval_bd_guidance.py

Question addressed. The importance-weight guidance of MASKED_GUIDANCE.pdf is stated for a fixed-length canvas. Block diffusion (BD3-LM style) generates *semi-autoregressively*: blocks are committed one at a time and length is emergent. Can the method be applied directly?

Answer: yes — the block-autoregressive factorization composes exactly with the per-block guided reverse process, provided the discriminator is defined on *partial paths*. This note gives the derivation, the two lemmas that make it work (prefix-constant cancellation; the bias of a block-marginal discriminator), and the estimator actually implemented.

1. Setup

Vocabulary: road vertices plus END; canvas form $x = [\text{dst}, \text{v}_0=\text{ori}, \text{v}_1, \dots]$. The BD model factorizes block-autoregressively over blocks of size B ,

$$p_\theta(x \mid od) = \prod_b p_\theta(x^b \mid x^{<b}, od), \quad (1)$$

each block factor realized by absorbing (masked) diffusion with SUBS $\hat{x}_0$ -prediction and first-hitting sampling. q denotes the target (exceptional-scenario) path distribution over the same canvases. We assume, as in the companion note, the same forward corruption for q and p_θ within a block, so $q(x_t^b \mid x_0^b) = p_\theta(x_t^b \mid x_0^b)$.

2. Block-conditional guided reverse step

Fix the committed prefix $x^{<b}$ and the current noised block x_t^b . The exceptional-target reverse posterior of the current block, conditioned on the prefix, expands exactly as in the fixed-canvas case but with all measures conditioned on $x^{<b}$:

$$q(x_{t-1}^b \mid x_t^b, x^{<b}) = \mathbb{E}_{x_0^b \sim p_\theta(x_0^b \mid x_t^b, x^{<b})} [w_b(x_0^b) \cdot q(x_{t-1}^b \mid x_t^b, x_0^b)] / \mathbb{E}[w_b], \quad (2)$$

$$w_b(x_0^b) = q(x_0^b \mid x^{<b}) / p_\theta(x_0^b \mid x^{<b}). \quad (3)$$

Eq. (2) is the exact conditional analogue of the companion note's Eq. (1); the forward-kernel ratio $q(x_t^b \mid x_0^b) / p_\theta(x_t^b \mid x_0^b) = 1$ cancels for the same reason (shared absorbing forward). The only new object is the **conditional** density ratio (3).

2.1 Lemma 1 (prefix-constant cancellation) — why a partial-path discriminator suffices

Write the conditional ratio via joint (partial-path) ratios:

$$w_b(x_0^b) = [q(x^{<b}, x_0^b) / p_\theta(x^{<b}, x_0^b)] \cdot [p_\theta(x^{<b}) / q(x^{<b})] = C^{-1} \cdot r(x^{<b} \parallel x_0^b), \quad (4)$$

where $r(u) = q(u) / p_\theta(u)$ is the ratio of **partial-path marginals** and $C = q(x^{<b}) / p_\theta(x^{<b})$ depends only on the (fixed) prefix — it is **constant across candidates**. The guided step (2) is self-normalized: substituting N candidates $x_0^{b(n)} \sim p_\theta(x_0^b \mid x_t^b, x^{<b})$ (mean-field, §3),

$$\hat{q}(x_{t-1}^b \mid \cdot) = \sum_n w_b^{(n)} q(x_{t-1}^b \mid x_t^b, x_0^{b(n)}) / \sum_n w_b^{(n)}, \quad (5)$$

any candidate-independent factor of w_b cancels between numerator and denominator. Hence **using $r(x^{<b} \parallel x_0^b)$ in place of w_b leaves (5) unchanged**: a discriminator trained to tell exceptional from reference *partial paths*, evaluated on [committed prefix $\parallel$ candidate completion], implements the conditional-ratio guidance exactly. By the Bayes-optimality identity, a partial-path discriminator D_ϕ trained with positives $\sim q$ -partials and negatives $\sim p_{\text{ref}}$ -partials satisfies $D/(1-D) \rightarrow q(u) / p_{\text{ref}}(u) = r(u)$ (with $p_{\text{ref}} = p_\theta$; see §4 on the negative class).

Finite-precision caveat. C cancels algebraically, but if the prefix is already strongly typed (C very large/small) the logit sits in the saturated region of D and candidate-level differences are crushed by finite precision. This motivates the bounded-logit head and label smoothing of §5 — saturation control matters even though C is "harmless" in exact arithmetic.

2.2 Lemma 2 (block-marginal discriminator is biased) — why the prefix must be included

Suppose instead the discriminator saw only the B-token block, estimating the **marginal** ratio $r_{\text{blk}}(x_0^b) = q_{\text{blk}}(x_0^b) / p_{\text{blk}}(x_0^b)$ where $q_{\text{blk}}(y) = \sum_u q(u)$ $q(y|u)$ marginalizes over prefixes u . Then

$$r_{\text{blk}}(y) / w_b(y) = [\sum_u q(u | y) \dots] \text{ — a function of } y, \text{ **not** a constant,} \quad (6)$$

because the prefix posterior $q(u | y)$ depends on the candidate y . The discrepancy does not cancel in (5); the resulting guidance targets a different (prefix-averaged) distribution. Intuitively: whether a 4-token fragment is "rerouting around a removed edge" is undecidable without the route that led into it. **The discriminator input must be the full partial path.**

3. First-hitting instantiation (what is implemented)

Within the current block, the guided reveal follows the companion note's Eqs. (3)–(5) with the conditional weight of §2. At each first-hitting step (one forward pass, one token revealed):

- **Proposals.** $N = 100$ candidates $x_0^{b,(n)}$ are drawn from the mean-field product of the SUBS marginals, $\hat{x}_0 = \prod_{\ell \in \text{masked}} p_\theta(x_0^\ell | x_t^b, x^{<b})$; revealed positions are copied through (frozen). An optional proposal temperature τ (companion §6 remedy) flattens the candidate distribution: $x_0 \sim \text{softmax}(\text{logits}/\tau)$.
- **Weights.** Each candidate is truncated at its first END (exclusive) / dst (inclusive) — partial paths never contain special tokens — padded with the dedicated PAD, and scored: $w^{(n)} = \exp(\gamma \cdot \text{logit}_\phi([x^{<b} \parallel x_0^{b,(n)}]))$, where $\exp(\text{logit}) = D/(1-D)$ and $\gamma \leq 1$ is an optional tempering exponent. Row-max subtraction before exponentiation (shift-invariance of (5)) gives numerical stability.
- **Reveal target.** $\bar{x}_0^\ell[k] \propto \sum_n w^{(n)} \mathbb{1}\{x_0^{(n),\ell} = k\}$; one uniformly-chosen masked position ℓ^* is revealed by sampling from $\bar{x}_0^{\ell^*}$ (absorbing posterior, companion Eq. 5). Zero-mass fallback: $\hat{x}_0$.
- **Diagnostics.** Per-step effective sample size $\text{ESS} = (\sum_n w^{(n)})^2 / \sum_n (w^{(n)})^2$ is logged. $\text{ESS} \approx N \Leftrightarrow$ flat weights $\Leftrightarrow$ guidance inert (either clone candidates — the proposal-entropy bottleneck of companion §6, observed $\text{ESS} \approx 9.93/10$ in the smoke run — or an uninformative discriminator). $\text{ESS} \approx 1 \Leftrightarrow$ weight collapse (variance).

3.1 Sequential composition and its bias

Across blocks the procedure is **sequential self-normalized importance sampling without resampling**: block b 's guided kernel treats the guided prefix as exact. Two approximations stack per block: (i) the self-normalization bias of (5), $O(1/N)$ for bounded weights (bounded by $e^{2\lambda\gamma}$ under the λ -bounded logit head); (ii) the mean-field proposal $\hat{p}$ replacing the true block posterior — identical in kind to the companion note's per-step approximation, not BD-specific. No further BD-specific approximation is introduced; in the $B \rightarrow$ canvas-length limit the scheme reduces to the fixed-canvas method, and at $B = 1$ to discriminator-reweighted autoregressive decoding.

4. The discriminator: training distribution = deployment distribution

Ingredient	Specification and rationale
Input	Canvas-form partial path $[\text{dst}, v_0, \dots, v_i]$ + scenario adjacency A_{scn} . Matches the guidance-time input exactly (Lemma 1).
Positives	Random-truncation partials of exceptional paths: j resampled every batch (multiple cuts per path across training — no truncation point is discarded; K cuts per path per epoch is equivalent in the limit). Data budget: 1% / 3% of except-0 (exp. 1) or of each of except-1...99 (exp. 2).
Negatives (denominator p_{ref})	data : normal-data partials (assumes $p_\theta = q_{\text{normal}}$). mix : 50% normal partials + 50% model samples — open-ended generations and one-shot mean-field block completions from the normal-trained BD model (<code>gen_bd_neg_pool.py</code>), i.e. the same distribution the candidates are drawn from at deployment. Rationale: the ratio's denominator is defined by the negative class; scoring mean-field candidates with a discriminator that has never seen off-manifold sequences yields arbitrary logits (the observed weight-variance pathology).
Adjacency channel	Per-transition features $[A_{\text{scn}}(v_{i-1}, v_i), \text{is-transition}, \text{deg}_{\text{scn}}(v_i)/\text{deg}_{\text{norm}}(v_i)]$ injected at every position, plus GraphSAGE-updated vertex embeddings under A_{scn} . The removed-edge signal is a direct table lookup — this is what should generalize to unseen scenarios (exp. 2), where each batch carries its own exception's adjacency.
Padding	Dedicated PAD token (= <code>n_vertex</code> , embedding-frozen) + masked mean pooling: the logit of a path is invariant to batch composition (fixes the vertex-0-padding pollution and the pooling dilution of the prior <code>Discriminator</code>).

5. Stabilizing the importance weights

- **Bounded logit head** $z \mapsto \lambda \cdot \tanh(z/\lambda)$ ($\lambda = 4$): $|\text{logit}| \leq \lambda$ structurally caps the weight ratio at $e^{2\lambda}$; the differentiable version of the clip-at-3 experiment in the prior project's logs.
- **Label smoothing** $\varepsilon = 0.05$ in the BCE: keeps D off $\{0,1\}$, bounding the Bayes-target odds and preventing saturation (see the caveat of §2.1).
- **Negative-class coverage** (the 'mix' ablation): addresses the out-of-support pathology at its source.
- **Weight tempering** $w \mapsto w^\gamma$, $\gamma \leq 1$: continuous variance–bias knob ($\gamma = 0$ recovers unguided sampling).
- **Self-normalization + row-max shift**: exact shift-invariance used for numerics; Lemma 1 guarantees the shift is harmless.
- **ESS logging** at every reveal: separates "guidance inert" ($\text{ESS} \approx N$) from "guidance unstable" ($\text{ESS} \approx 1$) — the two failure modes have opposite remedies (raise proposal entropy vs. temper weights).

6. Adjacency-masked proposals — Lemma 3 (exactness under the ideal discriminator)

Motivation (measured). The first experimental round found $\text{ESS} \approx 94\text{--}99$ out of $N = 100$ at every block size: mean-field candidates are near-clones, so reweighting has nothing to choose between (the companion note's §6 bottleneck). Raising the proposal temperature adds *isotropic* entropy — mass spreads over all 1,390 vertices, almost none of it on legal reroutes — and degrades arrival faster than it helps (measured: $\tau = 2$ costs -4 to -17% arrival for -2 to -6% removed-edge usage, ESS still ≈ 97). The fix is *directed* diversity: constrain the proposal to scenario-legal transitions.

6.1 The masked proposal

At a reveal step let M be the masked positions of the current block, and for each $\ell \in M$ let $p_\ell(\cdot)$ be the model's SUBS marginal (shared by all candidates — they condition on the same x_ℓ and prefix). Define the legality mask from the **revealed** neighbours only:

$$m_\ell(k) = A_{\text{scn}}[\text{left}(\ell), k]^{\mathbb{I}\{\text{left revealed, real vertex}\}} \cdot A_{\text{scn}}[k, \text{right}(\ell)]^{\mathbb{I}\{\text{right revealed, real vertex}\}}, \quad m_\ell(\text{END}) = m_\ell(\text{PAD}) = 1, \quad (8)$$

and the masked mean-field proposal $g(y) = \prod_{\ell \in M} p_\ell(y_\ell) m_\ell(y_\ell) / Z_\ell$, with $Z_\ell = \sum_k p_\ell(k) m_\ell(k)$ (fallback to unmasked p_ℓ if $Z_\ell \approx 0$). Because m depends only on revealed tokens — never on candidate-internal values — Z_ℓ is **identical across candidates**.

6.2 Lemma 3

Claim. Let $w(y) = q(y \mid x^{<b}) / p_\theta(y \mid x^{<b})$ be the exact conditional weight of Eq. (3) and f any statistic (e.g. the reveal indicator). If the target q assigns zero probability to paths using removed edges — true by construction, exceptional data are paths on the exceptional graph — then

$$\mathbb{E}_{y \sim \hat{p}}[w(y) f(y)] = \mathbb{E}_{y \sim g}[w(y) f(y)] \cdot \prod_\ell Z_\ell, \quad (9)$$

so the **self-normalized** estimator over g -candidates with **unchanged** weights w (no $\hat{p}/g$ correction) targets exactly the same guided reveal distribution as the $\hat{p}$ -proposal estimator.

Proof. Exact importance identity with proposal change: $\mathbb{E}_{\hat{p}}[w f] = \mathbb{E}_g[(\hat{p}/g) w f] + \mathbb{E}_{\hat{p}}[w f \mathbb{I}\{y \notin \text{supp } g\}]$. On $\text{supp } g$, $\hat{p}(y)/g(y) = \prod_\ell Z_\ell / m_\ell(y_\ell) = \prod_\ell Z_\ell$ (all $m = 1$ there), a candidate-independent constant. Off $\text{supp } g$, some position uses a transition absent from A_{scn} against a *revealed* neighbour; any completion y is then a path using a removed (or never-existing) edge, hence $q(y \mid x^{<b}) = 0$ and $w(y) = 0$ — the excluded term vanishes *identically*, not approximately. Substituting into the self-normalized ratio, $\prod_\ell Z_\ell$ cancels between numerator and denominator (as C did in Lemma 1). ■

Reading. Under the ideal discriminator, adjacency masking discards only candidates whose weight is exactly zero: it is a **Rao-Blackwellization** of the same estimator — all N samples land in the region where the target lives, which is precisely a variance reduction, never a bias. With a *learned* D the difference between the two estimators is bounded by the (spurious) weight mass $D/(1-D)$ that the discriminator assigns to illegal candidates — the masked estimator simply deletes discriminator error on the illegal region.

6.3 Scope and caveats

- **Revealed-neighbour masking only.** If masking instead chained through candidate-internal values (sequential within-candidate sampling), Z would depend on the candidate and the $\hat{p}/g$ correction would no longer be constant — the cancellation is specific to the position-independent mask (8). Consequence: two *adjacent masked* positions can still be mutually illegal inside one candidate; the D-weights (and the next reveal's updated mask) handle the residue.
- **Legality of special tokens.** END/PAD are always legal (ending is not a traversal); MASK is already excluded by SUBS. The *dst* prefix (position 0) is not a traversal and imposes no constraint.

- **Unguided control.** Masking without the discriminator (uniform weights over g-candidates $\Leftrightarrow$ revealing directly from the masked marginal) isolates how much improvement comes from the constraint itself vs. from reweighting among legal reroutes — included in the experiment grid as `adj-only`.
- **Zero-support fallback.** If no token is legal under (8) ($Z_\ell \approx 0$, e.g. a dead-end vertex in A_{scn}), the position falls back to the unmasked marginal; such reveals are the analogue of P1's unfixable pairs (0 observed so far).

7. Experimental design

Axis	Values
Backbone (normal-trained, reused)	mask kernel, blk $\in \{4, 8, 16, 32, 64\}$ (main axis, fam 0.05 / 1% / exp. 1); pivots blk {4, 16} for the other axes
Scenario family	v4-0.05 (200 edges removed), v4-0.1 (both share the v3 normal graph — verified identical; excepts are edge-subsets)
Data budget	1%, 3% (except-0 rows for evaluation are reserved before the training split; seed 777)
Experiments	1: disc on except-0 $\rightarrow$ eval except-0. 2: disc on except-1...99 (1% each, per-batch scenario + adjacency) $\rightarrow$ eval except-0 (unseen-scenario generalization)
Disc negatives	data / mix (ablation)
Proposal temperature	$\tau \in \{1, 1.5, 2\}$ ($\tau > 1$ on blk {4, 16}, main disc — companion §6 remedy; measured: hurts more than it helps)
Adjacency-masked proposals (§6)	blk {4, 8, 16, 32, 64} $\times$ {adj-only (no disc), adj+guided (mix disc)} on fam 0.05 / 1% / exp. 1; blk {4, 16} $\times$ adj+guided for fam 0.1 and for the exp. 2 disc
Estimator	N = 100 importance samples, $\gamma = 1$, guidance at every reveal (no t-window: first-hitting reveals are irreversible, so any unguided reveal locks in its error — the old $t < 10$ heuristic does not transfer)
Evaluation	1,000 reserved except-0 OD pairs; arrival / validity vs. A_{except0} / EM / PC / removed-edge usage (path-, per-path-, edge-rate) / len corr / ESS; raw (no post-processing)

Runs: 1 negative pool + 8 discriminator trainings + 7 unguided baselines + 26 guided evaluations (22 main/pivot + 4 temperature), orchestrated by `run_bd_guidance_all.sh` in tmux over 4 GPUs.